instrumentation.py — technical

`src/moe_pruning/model/instrumentation.py · 85 lines · moe_pruning.model.instrumentation`

Purpose

Consistency and safety checks over forward records.

Formulas

```
expected = sum_k gate[..., k] * expert_output[..., k, :]; max_rel_error = max|expected - moe_output| / max|moe_output|
```

Step by step

Classes and functions in source order. Signatures, line ranges, raised exceptions and settings are extracted from the code; numbered steps describe what the code does.

`reconstruct_moe_output`

```
def reconstruct_moe_output(record: MoEForwardRecord) -> torch.Tensor
```

lines 10-12

Recompute sum_k gate_k * E_k(x) from the recorded routing and expert outputs.

`gate_consistency_error`

```
def gate_consistency_error(record: MoEForwardRecord) -> dict[str, float]
```

lines 15-29

Compare the layer's real output with sum_k g_k * E_k(x).

`class RecordCollector`

lines 32-56

Callback that keeps records in memory (for small probes and tests only).

`RecordCollector.__call__`

```
def __call__(self, record: MoEForwardRecord) -> None
```

lines 39-50

Store one record (moved to CPU unless to_cpu=False).

1. Optionally copy every tensor of the record to CPU, then append.

`RecordCollector.by_layer`

```
def by_layer(self) -> dict[str, list[MoEForwardRecord]]
```

lines 52-56

Records grouped by layer id.

`gate_consistency_tolerance`

```
def gate_consistency_tolerance(dtype: torch.dtype) -> float
```

lines 59-68

Relative tolerance for `gate_consistency_error` at the model's precision.

1. 1e-4 for float32; 3e-2 for float16/bfloat16 (rounding of recomputed outputs).

`class NonFiniteError`

lines 71-72 · bases FloatingPointError

Raised when inf/NaN values appear (typically float16 overflow).

`assert_finite`

```
def assert_finite(tensor: torch.Tensor, what: str, dtype: torch.dtype | None=None) -> None
```

lines 75-85 · raises NonFiniteError

Raise a clear error on inf/NaN (typical for float16 overflow, e.g. bf16-trained models on T4).

1. Return if all values are finite; otherwise raise `NonFiniteError`, adding a float16-range explanation when the model dtype is float16.

Errors raised

Function	Exception	Message (f-string fields in braces)
`assert_finite`	NonFiniteError	Non-finite values (inf/NaN) in {what}.{hint}